Classical Field Theory — Exercise Sheet 14

First-order systems and real structures

Solutions

All spinor entries commute. Use Lecture 14's Hermitian inner product, unitary Fourier convention and real form $\Omega(h,g) = 2\,\mathrm{Im}\int h^\dagger g$. Problems 1 and 3 supply new four-component matrices; Problem 4 returns to the actual two-component matrices of Lecture 14.

Exercise 14.1 (Four components in three space dimensions). Let I_2 be the two-by-two identity and put

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \Gamma^0 = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix}, \quad \Gamma^i = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix}.$$

Define $\beta_D = \Gamma^0$, $\alpha_D^i = \Gamma^0\Gamma^i$ and $H_D(k) = \alpha_D^i k_i + m\beta_D$, where $m \geq 0$.

(a) Verify every Clifford anticommutator for signature $(+,-,-,-)$. Prove that $H_D(k)$ is Hermitian and find its eigenvalues, multiplicities and orthogonal spectral projectors, including the exceptional $m = k = 0$ case.

(b) For $\mathrm{i}\psi_t = (-\mathrm{i}\alpha_D^i\partial_i + m\beta_D)\psi$, derive the conserved current $\rho = \psi^\dagger\psi$, $j^i = \psi^\dagger\alpha_D^i\psi$ and prove $|n\cdot j| \leq \rho$ for every unit vector n. Explain why verifying these matrices alone does not prove the spin representation or Lorentz-covariance theorem cited in Lecture 14.

Solution. For the first part, direct multiplication gives $\sigma_i\sigma_j + \sigma_j\sigma_i = 2\delta_{ij}I_2$. The block products therefore give

$$(\Gamma^0)^2 = I_4, \quad \Gamma^0\Gamma^i + \Gamma^i\Gamma^0 = 0, \quad \Gamma^i\Gamma^j + \Gamma^j\Gamma^i = -2\delta_{ij}I_4, \qquad \alpha_D^i = \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix}.$$

Both α_D^i and β_D are Hermitian. Their anticommutators give

$$H_D^2 = (|k|^2 + m^2)I_4, \qquad \mathrm{tr}\, H_D = 0.$$

For $\nu > 0$, the Hermitian spectral theorem permits only eigenvalues $\pm\nu$; trace zero gives two of each. The projectors

$$P_\pm = \frac{1}{2}(I_4 \pm H_D/\nu)$$

are Hermitian, square to themselves, sum to the identity and have trace two. At $m = k = 0$, the zero matrix has a four-dimensional zero eigenspace; the displayed division by ν is not used there.

For the second part, substitute the evolution and its adjoint into the product derivative. The two mass terms cancel and give

$$\partial_t(\psi^\dagger\psi) = -\partial_i(\psi^\dagger\alpha_D^i\psi).$$

For $A_n = n_i\alpha_D^i$, Hermiticity and $A_n^2 = I_4$ imply $|\psi^\dagger A_n\psi| \leq |\psi|\,|A_n\psi| = |\psi|^2$. This proves the claimed bound. The calculation verifies a Clifford representation and a current. It does not construct the action of the spin group on Lorentz transformations, which is an additional theorem.

Exercise 14.2 (A constant symmetric-hyperbolic system). Let $A^1, \dots, A^d$ be constant Hermitian N-by-N matrices and B a constant skew-Hermitian matrix. For complex columns consider

$$w_t + A^i \partial_i w + Bw = 0, \qquad c_* = \sup_{|n|=1} \|n_i A^i\|_{\mathrm{op}}.$$

Here the operator norm is the Euclidean matrix norm.

(a) Construct the unique distributional solution in $C(\mathbb{R}; L^2)$ for each L^2 datum and prove norm preservation. Derive the smooth local current.

(b) Using Lecture 5's moving-ball argument, prove that data supported in a closed set K remain supported in $\{x : \mathrm{dist}(x, K) \le c_*|t|\}$. Give the limiting argument for rough data and include $c_* = 0$.

(c) For $d = N = 2$, take $A^1 = \sigma_1$, $A^2 = 2\sigma_3$ and $B = \mathrm{i}\sigma_2$. Compute c_* and exhibit a column and normal attaining the current bound. Would Hermitian $B = I_2$ preserve the same norm?

Solution. For the first part, put

$$\widehat{w}(t, k) = e^{-t(\mathrm{i}k_i A^i + B)} \widehat{w}(0, k).$$

The exponential is unitary at real t, k. It is pointwise continuous in t, so dominated convergence and Plancherel give a norm-preserving $C(\mathbb{R}; L^2)$ evolution. Smooth frequency cutoffs give bounded-frequency solutions; their uniform L^2 convergence on compact time intervals passes the differential equation to compact spacetime tests. For uniqueness, cut off a zero-data difference in frequency. Its bounded generator makes the distributional time equation a norm-primitive equation, as in Lecture 14. Its squared norm has zero derivative, so it vanishes. Remove the cutoff to obtain uniqueness. For a smooth solution, Hermiticity of the A^i and $B^\dagger = -B$ give

$$\partial_t \rho + \partial_i j^i = 0, \qquad |j \cdot n| = |w^\dagger (n_i A^i) w| \le c_* \rho.$$

The skew lower-order term cancels rather than adding damping.

For the second part, first take smooth compact initial data; the Fourier formula gives a smooth solution since every differentiated integrand is bounded by a polynomial times the initial Schwartz profile. For $c_* > 0$, let $E(t)$ be its charge on $B_{R - c_* t}(a)$, $0 \le t < R/c_*$. Polar-coordinate FTC and Lecture 5's ball divergence formula give

$$E'(t) = -\int_{\partial B_{R - c_* t}(a)} (j \cdot n + c_* \rho) \mathrm{d}S \le 0.$$

Smooth compact data are dense in L^2. Unitarity gives convergence of the data and evolved fields in L^2; hence their squared norms on these balls converge. The same inequality holds for arbitrary data. If $\mathrm{dist}(a, K) > c_* t$, choose $c_* t < R < \mathrm{dist}(a, K)$: the initial ball charge is zero, so the evolved field vanishes on a neighbourhood of a. A countable cover proves the almost-everywhere support statement. For $c_* = 0$, every A^i is zero (take $n = e_i$); evolution is the pointwise unitary matrix e^{-tB} and preserves support exactly. Reversing time proves the negative-time assertion.

For the last part, the Pauli anticommutators give

$$(n_1 A^1 + n_2 A^2)^2 = (n_1^2 + 4n_2^2) I_2.$$

On the unit circle its norm is at most two, with equality at $n = e_2$. For $w = e_1$ one has $\rho = 1$ and $j \cdot e_2 = 2$. Replacing B by I_2 instead gives $\partial_t \rho + \partial_i j^i = -2\rho$; its Fourier formula contains e^{-t} times a unitary matrix, so the squared norm is multiplied by e^{-2t}.

Exercise 14.3 (Weyl sectors and a mass coupling). Use the four-component matrices of Problem 1. Define $\Gamma^5 = \mathrm{i}\Gamma^0\Gamma^1\Gamma^2\Gamma^3$ and $P_{\mathrm{ch},\pm} = (I_4 \pm \Gamma^5)/2$. Compute Γ^5 and its two eigenspaces. Write the Dirac Hamiltonian in the coordinates

$$\psi = \frac{1}{\sqrt{2}}\begin{pmatrix} u+v \\ u-v \end{pmatrix}, \qquad u, v \in \mathbb{C}^2.$$

Determine when the two chiral subspaces are invariant under evolution, and derive the two Weyl equations in the massless case.

Solution. Multiplication of the four blocks gives

$$\Gamma^5 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}.$$

It is Hermitian and squares to I_4. Its positive and negative spaces consist of columns $(u,u)^T/\sqrt{2}$ and $(v,-v)^T/\sqrt{2}$, respectively. The associated projectors are $P_{\mathrm{ch},\pm}$ as defined in the question. For the unitary matrix

$$W = \frac{1}{\sqrt{2}}\begin{pmatrix} I_2 & I_2 \\ I_2 & -I_2 \end{pmatrix},$$

block multiplication yields

$$W^\dagger \Gamma^5 W = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix}, \qquad W^\dagger H_D(k) W = \begin{pmatrix} \sigma\cdot k & mI_2 \\ mI_2 & -\sigma\cdot k \end{pmatrix}.$$

Thus in position space

$$\mathrm{i}u_t = -\mathrm{i}\sigma\cdot\nabla u + mv, \qquad \mathrm{i}v_t = +\mathrm{i}\sigma\cdot\nabla v + mu.$$

At zero mass these are independent two-component Weyl equations. At positive mass a nonzero column in either sector produces an immediate component in the other. No independent massive evolution on a single chiral sector follows by discarding that equation.

Exercise 14.4 (A commuting Majorana-type real subspace). Return to Lecture 14's two-component matrices $\alpha = -\sigma_3$, $\beta = \sigma_1$ and define the real-linear, antilinear-in-complex-scalars map

$$C\psi = \sigma_3\overline{\psi}.$$

Prove $C^2 = I$, characterize its fixed space in L^2, and show that Dirac evolution preserves it. Derive the evolution of its two real entries. Restrict the real form Ω to that fixed space. Is the result nondegenerate? Explain why this calculation does not give a classical commuting replacement for a quantum Majorana field or its statistics.

Solution. Conjugating twice gives $C^2 = \sigma_3\overline{\sigma_3} = I$. The fixed condition says that the first entry is real and the second is purely imaginary. Write it as $\psi = (a, \mathrm{i}b)^T$. In the position-space evolution

$$\psi_t = -\alpha\psi_x - \mathrm{i}m\beta\psi,$$

conjugation reverses the sign of i, while anticommutation reverses the mass matrix sign again. Commutation of σ_3 with α then shows that $C\psi$ solves the same equation. The L^2 uniqueness theorem proves invariance, also for rough data. Substitution gives the real distributional equations

$$a_t = a_x + mb, \qquad b_t = -b_x - ma.$$

For two fixed tangent columns $h = (a, \mathrm{i}b)^T$ and $g = (c, \mathrm{i}d)^T$, their inner product integrand is $ac + bd$, which is real. Hence

$$\Omega(h, g) = 2\,\mathrm{Im} \int (ac + bd)\mathrm{d}x = 0.$$

There are nonzero fixed columns, so the restriction is degenerate. Multiplication by i does not preserve this fixed space, and Lecture 14's test $\Omega(h, \mathrm{i}h) = 2\|h\|_2^2$ is not an internal test here. The present example concerns commuting classical entries and a real subspace of their evolution. Quantum fermionic variables and statistics are additional structures, not consequences of this restriction.